$$\mathrm{CaC_2(s)} + 5\mathrm{H_2O(l)} \rightarrow \mathrm{CaO(s)} + 2\mathrm{CO_2(g)} + 5\mathrm{H_2(g)},\ \Delta\hat{H}_r^\circ = 69.36\ \text{kJ/kmol}$$

a. Endothermic. The reactor will have to be heated to keep the temperature constant. The temperature would decrease under adiabatic conditions. The energy required to break the reactant bonds is more than the energy released when the product bonds are formed.

b.

$$\Delta\hat{U}_r^\circ = \Delta\hat{H}_r^\circ - RT\left[\sum_{\substack{\text{gaseous}\\\text{products}}} \nu_i - \sum_{\substack{\text{gaseous}\\\text{reactants}}} \nu_i\right] = 69.36\frac{\text{kJ}}{\text{mol}} - \frac{8.314\ \text{J}}{\text{mol}\cdot\text{K}}\left|\frac{1\ \text{kJ}}{10^3\ \text{J}}\right|\frac{298\ \text{K}}{}\left|\frac{(7-0)}{}\right.$$

$$= \underline{\underline{52.0\ \text{kJ/mol}}}$$

$\Delta\hat{U}_r^\circ$ is the change in internal energy when 1 g-mole of $\mathrm{CaC_2(s)}$ and 5 g-moles of $\mathrm{H_2O(l)}$ at 25°C and 1 atm react to form 1 g-mole of CaO(s), 2 g-moles of $\mathrm{CO_2(g)}$ and 5 g-moles of $\mathrm{H_2(g)}$ at 25°C and 1 atm.

c.

$$Q = \Delta U = \frac{n_{\mathrm{CaC_2}}\Delta\hat{U}_r^\circ}{\nu_{\mathrm{CaC_2}}} = \frac{150\ \text{g CaC}_2}{}\left|\frac{1\ \text{mol}}{64.10\ \text{g}}\right|\frac{52.0\ \text{kJ}}{1\ \text{mol CaC}_2} = \underline{\underline{121.7\ \text{kJ}}}$$

Heat must be transferred to the reactor.

a. $N_2(g)+O_2(g)\rightarrow 2NO(g)$, $\Delta\hat{H}_r^o = 2\left(\Delta\hat{H}_f^o\right)_{NO(g)} = 2\left(\overset{\text{Table B.1}}{\overset{\downarrow}{90.37\ \text{kJ/mol}}}\right) = \underline{\underline{180.74\ \text{kJ/mol}}}$

b. $n-C_5H_{12}(g)+\frac{11}{2}O_2(g)\rightarrow 5CO(g)+6H_2O(l)$

$$\Delta\hat{H}_r^o = 5\left(\Delta\hat{H}_f^o\right)_{CO(g)} + 6\left(\Delta\hat{H}_f^o\right)_{H_2O(l)} - \left(\Delta\hat{H}_f^o\right)_{n-C_5H_{12}(g)}$$

$$=\left[(5)(-110.52)+(6)(-285.84)-(-146.4)\right]\ \text{kJ/mol} = \underline{\underline{-2121.2\ \text{kJ/mol}}}$$

c. $C_6H_{14}(l)+\frac{19}{2}O_2(g)\rightarrow 6CO_2(g)+7H_2O(g)$

$$\Delta\hat{H}_r^o = 6\left(\Delta\hat{H}_f^o\right)_{CO_2} + 7\left(\Delta\hat{H}_f^o\right)_{H_2O(g)} - \left(\Delta\hat{H}_f^o\right)_{C_6H_{14}(l)}$$

$$=\left[(6)(-393.5)+7(-241.83)-(-198.8)\right]\ \text{kJ/mol} = \underline{\underline{-3855\ \text{kJ/mol}}}$$

d. $Na_2SO_4(l)+4CO(g)\rightarrow Na_2S(l)+4CO_2(g)$

$$\Delta\hat{H}_r^o = \left(\Delta\hat{H}_f^o\right)_{Na_2S(l)} + 4\left(\Delta\hat{H}_f^o\right)_{CO_2(g)} - \left(\Delta\hat{H}_f^o\right)_{Na_2SO_4(l)} - 4\left(\Delta\hat{H}_f^o\right)_{CO(g)}$$

$$=\left[(-373.2+6.7)+(4)(-393.5)-(-1384.5+24.3)-4(-110.52\right]\ \text{kJ/mol} = \underline{\underline{-138.2\ \text{kJ/mol}}}$$